Kishor Jothimurugan

kishor1996@gmail.com • keyshor.github.io

Education

University of Pennsylvania **Philadelphia, USA**
Ph.D. in Computer and Information Science, GPA 4/4 *2017–2023*
Thesis Topic: Specification-Guided Reinforcement Learning
Committee: Dr. Osbert Bastani, Dr. Thomas A. Henzinger, Dr. Sampath Kannan, and Dr. George J. Pappas
Advised by Dr. Rajeev Alur

Chennai Mathematical Institute **Chennai, India**
B.Sc. (Honors) in Mathematics and Computer Science, CGPA 9.77/10 *2014–2017*
Ranked among top 3 students

Work Experience

Two Sigma Investments, LP 2023–Current
Quantitative Researcher
Systematic macro research

Amazon Web Services Summer 2022
Applied Scientist Intern, AI Labs
Incorporating program semantics in transformer-based code generation models
Mentors: Dr. Nathan Fulton, Dr. Siddhartha Jain and Dr. Baishakhi Ray

Nokia Bell Labs Summer 2020
Research Intern, Data and AI Lab
An application of deep RL to regenerative stopping problems
Mentors: Dr. Matthew Andrews, Dr. Jeongran Lee and Dr. Lorenzo Maggi

Amazon Web Services Summer 2019
Software Development Engineering Intern, Automated Reasoning Group
Using machine learning to improve usability of taint analysis
Mentors: Dr. Andrew Gacek and Dr. Lee Pike

ENS Paris-Saclay Summer 2017
Research Intern, Formal Methods Group
Mentor: Dr. Dietmar Berwanger

Publications

First (co)author in all publications; names are listed alphabetically in some papers
* equal contribution, † authors in alphabetical order

Journal Publications

[CACM 26] Specification-Guided Reinforcement Learning
R. Alur, S. Bansal, O. Bastani and *K. Jothimurugan*†
Communications of the ACM, 2026

[PSD 22] A Framework for Transforming Specifications in Reinforcement Learning
R. Alur, S. Bansal, O. Bastani and *K. Jothimurugan*†
Principles of Systems Design: Essays Dedicated to Thomas A. Henzinger on the Occasion of His 60th Birthday (Invited), 2022

[TECS 21] Compositional Learning and Verification of Neural Network Controllers
R. Ivanov*, *K. Jothimurugan**, S. Hsu, S. Vaidya, R. Alur and O. Bastani
ACM Transactions on Embedded Computing Systems, 2021

Conference Publications

[NeuS 25] Specification-Guided Reinforcement Learning
K. Jothimurugan, S. Bansal, O. Bastani and R. Alur
International Conference on Neuro-symbolic Systems, 2025

[ICML 23] Robust Subtask Learning for Compositional Generalization
K. Jothimurugan, S. Hsu, O. Bastani and R. Alur
International Conference on Machine Learning, 2023
Acceptance rate: 27.9%

[CAV 23] Policy Synthesis and Reinforcement Learning for Discounted LTL
R. Alur, O. Bastani, *K. Jothimurugan*†, M. Perez, F. Somenzi and A. Trivedi
International Conference on Computer Aided Verification, 2023
Acceptance rate: 26%

[CAV 22] Specification-Guided Learning of Nash Equilibria with High Social Welfare
K. Jothimurugan, S. Bansal, O. Bastani and R. Alur
International Conference on Computer Aided Verification, 2022
Acceptance rate: 24.4%

[NeurIPS 21] Compositional Reinforcement Learning from Logical Specifications
K. Jothimurugan, S. Bansal, O. Bastani and R. Alur
Advances in Neural Information Processing Systems, 2021
Acceptance rate: 25.7%

[AISTATS 21] Abstract Value Iteration for Hierarchical Reinforcement Learning
K. Jothimurugan, O. Bastani and R. Alur
International Conference on Artificial Intelligence and Statistics, 2021
Acceptance rate: 29.8%

[LICS 20] Space-efficient Query Evaluation over Probabilistic Event Streams
R. Alur, Y. Chen, *K. Jothimurugan*† and S. Khanna
Symposium on Logic in Computer Science, 2020
Acceptance rate: 39.6%

[NeurIPS 19] A Composable Specification Language for Reinforcement Learning Tasks

K. Jothimurugan, R. Alur and O. Bastani
Advances in Neural Information Processing Systems, 2019
Acceptance rate: 21.1%

Preprints and Technical Reports

Learning Algorithms for Regenerative Stopping Problems with Logistics Applications
K. Jothimurugan, M. Andrews, J. Lee and L. Maggi
Nokia Bell Labs Intern Report

Tutorials

[AAAI 23] Specification-Guided Reinforcement Learning
Presenters: *K. Jothimurugan*, S. Bansal, R. Alur and O. Bastani
AAAI Conference on Artificial Intelligence, 2023
A comprehensive tutorial on RL algorithms for learning control policies from logical specifications

Open Source Tools

DIRL — Github link
Compositional RL algorithm for learning from temporal specifications

HIGH-NASH — Github link (CAV Artifact Evaluation Badge: Functional)
Multi-agent RL algorithm for learning Nash equilibria with high social welfare

SARL — Github link
Hierarchical RL algorithm leveraging user provided state abstractions

SPECTRL — Github link
Generating shaped rewards from temporal specifications

Teaching Experience

Guest Lecturer

Computer-Aided Verification, CIS 673 (Class of ~ 20) *Fall 2021*
Graduate Course, University of Pennsylvania
Topic: Techniques for Verifying Robustness of Neural Networks

Teaching Assistant

Principles of Embedded Systems, CIS 540 (Class of ~ 20) *Spring 2019*
Graduate Course, University of Pennsylvania

Automata, Computability and Complexity, CIS 262 (Class of ~ 150) *Fall 2018*
Undergraduate Course, University of Pennsylvania

Discrete Mathematics (Class of ~ 30) *Spring 2017*
Undergraduate Course, Chennai Mathematical Institute

Design and Analysis of Algorithms *Fall 2016*
Online Course offered by Indian Institute of Technology, Madras

Mentoring and Outreach

Graduate Student Mentoring 2020–2022
Steve Hsu, Masters Student, University of Pennsylvania
Topic: Compositional reinforcement learning for multi-task generalization
Co-authored two papers

Penn CIS Mentorship Program 2022
Participated as a mentor
Department-wide effort to connect early career PhD students with senior PhD students
Attracted participation from 34 first-year students

Awards

CTL Teaching Certificate *Spring 2022*
Awarded by Center for Teaching and Learning, University of Pennsylvania

CMI Undergraduate Scholarship *2014–2017*
Awarded by CMI to undergraduate students for excellence in academics

Research Presentations

Invited Talks

IARCS Verification Seminar Series *Summer 2023*
Title: Specification-Guided Reinforcement Learning

University of Illinois Urbana-Champaign *Spring 2023*
Title: Specification-Guided Reinforcement Learning

University of Virginia *Spring 2023*
Title: Specification-Guided Reinforcement Learning

NYU Formal Methods Group *Fall 2022*
Title: Specification-Guided Reinforcement Learning

Microsoft PROSE Group *Fall 2022*
Title: Specification-Guided Reinforcement Learning

Thomas Henzinger's Group, IST Austria *Fall 2021*
Title: Reinforcement Learning from Logical Specifications

Simons Institute, UC Berkeley *Spring 2021*
Workshop on Games and Equilibria in System Design and Analysis
Title: Abstract Value Iteration for Hierarchical Reinforcement Learning

Presentations and Posters (from publications)

Conferences: NeurIPS 2019 and 2021, LICS 2020, AISTATS 2021, EMSOFT 2021, CAV 2022, ICML 2023.

Workshops: DeepRL @ NeurIPS 2020 and 2022, SafeRL @ NeurIPS 2021, SYNT @ CAV 2022.

Review Service

Conferences: NeurIPS 2022, AAAI 2023, AISTATS 2023, ESOP 2023, L4DC 2023, ICML 2023, CAV 2023, NeurIPS 2023, ICLR 2024, ICML 2024, NeurIPS 2024, AAAI 2025, ICML 2025.

Journals: IEEE TCAD, JAIR.